Work Sheet – 08

Course :– Aptitude **Topic :- Series , Analogy, Classifications**

1. Find the missing number in the following series 226, 213, 202, 195, 190, - -, 185
   a) 165 b)175 c)187 d)178
   Sol:- $226-213=13, 213-202=11\ 202-195=7, 195-190=5$
   the differences are consecutive prime numbers. so the next number differ by 3
   Missing number is 190-3=187

2. In the following series a wrong number is given. find the wrong number
   2,3,6,9,36,140,246,253
   a)246 b)140 c)187 d)128
   Sol: -
   $$2+1=3, 3(2)=6$$
   $$6+3=9, 9(4)=36$$
   $$36+5=41, 41(6)=246$$
   natural numbers are added and multiplied consecutively So the wrong number is 140

3. what is the value of (d) in the new series if the series have the same pattern for series
   4,14,31,95,193 7,a,b,c,d
   a) 413 b)457 c)317 d)301
   Sol:- $4(3)+2=14, 14(2)+3=31$
   Series is formed by alternate multiplication and addition with 2 and 3
   So the new series is $7(3)+2=23=a\ 23(2)+3=49=b, c=\ 49(3)+2$
   $=149\ , 149(2)+3=301=d$

4. 11,28,327,464,?
   a)2413 b)5125 c)6317 d)7301
   Sol :- It is a different type of series We write
   $$11=\frac{1}{1^3}$$
   $$28=\frac{2}{2^3}$$
   $$327=\frac{3}{3^3}$$
   so a number and its cube are grouped to form a number So next number is
   $\left(\frac{5}{5^3}=5125\right)$

5. 2,6,3,5,35,7,11,143,13,--,-,-
   a)12,156,13 b)14,210,15 c)16,272,17 d)17,323,19
   Sol :- 2(3) = 6, 5(7) = 35

product of consecutive prime numbers give the middle number So the next numbers are 17,323,19

6. Find the wrong number in 1,2,8,48,384,3456

a)2 b)3456 c)384 d)48

Sol:- $1(2) = 2,\quad 2(4) = 8,\quad 8(6) = 48,\quad 48(8) = 384$

So Previous term is multiplied with consecutive even numbers to form series

Next number is 384(10) = 3840 so wrong number is 3456

7. Find the wrong one in the series: 5,8,15,26,50,98,194

a)15 b)26 c)194 d)98

Sol:- $(5-1)\times 2 = 8$

$(8-1)\times 2 = 14$

$(14-1)\times 2 = 26$ so Wrong number is 15

8. Find the wrong one : 6,12,36,80,150,252,392

a)80 b)6 c)150 d)36

Sol :- $(1^2 + 1^3 = 2)$,

$(2^2 + 2^3 = 12)$,

$(3^2 + 3^3 = 36)$ so 6 is the wrong one

9. 5,12,17,29,46,75,121,?

a)122 b)144 c)150 d)196

Sol:- 5 + 12 = 17, 12 + 17 = 29

Consecutive numbers are added to give next number

So the next number is 75 + 121 = 196

10. 5, 11, 23, 47, ?, 191

a)122 b)144 c)95 d)24

Sol:- $6\times 2 + 3 = 15,\ 15\times 2 + 5 = 35\ ,35\times 2 + 7 = 77$

each number is doubled and added to consecutive odd numbers so the next number is

$77 \times 2 + 9 =$ **163**

Directions :- In this type of questions, three words are given. In these words the first words are related to each other in some way. You are required to decide the relationship of the third word with options given on the basis of the first two given words.

11. India: New Delhi:: Pakistan: ?

A) Lahore B) Islamabad C) Peshawar D) Hyderabad

Sol:-India's Capital city is New Delhi

Pakistan's Capital city is Islamabad

12. Flower: Bud:: Fruit:?
A) Taste B) Seed C) Flower D) Twing
Sol :-Flowers come from Bud Fruits come from Flower

13. Elbow: Wrist: Knee ?
A) Ankle B) Fingers C) Foot D) Toes
Sol:- Elbow and Wrist are joints in the arm Ankle and Knee are joints in the Leg

14. Cow: Butter:: Mustard: ?
A) Flower B) Oil C) Plant D) Seed
Sol:- We get Butter from Cow's Milk We get Oil from Mustard

15. Food: Stomach:: Fuel: ?
A) Automobile B) Engine C) Truck D) Plane
Sol:- Food gives energy to the Stomach and Fuel provides energy to the engine

16. Church: England:: Kirk:?
A) Scotland B) Cathedral C) Cork D) Bishop
Sol:- Kirk is often used specifically for the Church of Scotland

17. Grapes: Wine:: Wheat:?
A) Grain B) Maize C) Bread D) Field
Sol:- Grapes are used to make Wine
Wheat is used to make bread

18. Seismograph: Earth quakes:: Taseometer: ?
A) Volcanoes B) Resistances C) Leiond Slides D) Strains
Sol:- Seismometer is used to record the motion of the ground during an Earthquake
Taseometer is instrument for measuring physical strain in a structure

19. Funk: Vitamins:: Curie: ?
A) Uranium B) Radium C) Radio activity D) Photography
Sol:- Funk discovered Vitamins and
Curie discovered Radium

20. Atom: Matter:: Particle: ?
A) Proton B) Electron C) Molecule D) Dust
Sol:- The smallest unit of Matter is Atom and That of dust is particle.

Directions (26-35): In each of the following questions, four pairs of words are given out of the words in three pairs dear a certain common relations. Choose the pair in which the words are differently related.

21. A) Stamp: Letter B) Ticket: Train C) Ink: Pen D) Car: Engine
Ans: Car: Engine (4)

Except 4, the first one is required for the performance of the second in all the other options.

22. A) Death: Disease B) Milk : Butter C) Grape : Wine D) Water : Oxygen
Ans: Death: Disease (1)
Except 1, the first one is required for the happening of the second in all the other options.

23. A) Hard: Soft B) Long: High C) Sweet: Sour D) Pointed: Blunt
Ans: Long: High (2)
Except this, remaining are opposites to one another.

24. A) Broad: Wide B) Light: Heavy C) Tiny: Small D) Big: Large
Ans: Light: Heavy (2)
Except this, remaining are synonyms to each other.

25. A) Petrol: Car B) Ink: Pen C) Garbage: Dustbin D) Car: Wheel
Ans: Car: wheel (4)
Except (4) in all the other three, second one contains the first one.

26. A) Army: General B) Team: Captain C) Creche: Infant D) Meeting: Chairman
Ans: Creche: Infant (3)
Except 3, in all the other pairs, second is the head of the first.

27. A) Shirt: Dress B) Boy: Girl C) Mango: Fruit D) Table: Furniture
. Ans: Boy: Girl (2)
Except 2, all the others are types of one another.

28. A) Newspaper: Editor B) Film: Director C) Stamps: Philatelist D) Book: Author
Ans: Stamps: Philatelist (3)
Except (3), in all the other pairs, first is prepared by the second.

29. A) Flower: Petal B) Chair: Leg C) Circle: Arc D) Cover: Page
Ans: Except 4, in all other pairs, second one is the part of the first one

30. A) Book: Page B) Table: Drawer C) Loom: Cloth D) Car: Wheel
Ans: Loom : Cloth (3)
Except 3,in all the other pairs, second one is the part of the first one

PRACTICE QUESTIONS

1. 13,16,20,25,31,?
   a) 33 b)38 c)39 d) 40

2. 129,120,112,105,99,?
   a) 91 b)94 c)95 d)93

3. 4,12,48,240,1440,?
   a) 5832 b)4096 c)4340 d) 10080

4. 6,12,24,48,96,?
   a) 126 b)142 c)192 d)196

5. 32,156,182,210,240,?
   a) 252 b)441 c)144 d)385

6. 21,169,225,289,361,?
   a) 343 b)441 c)144 d)385

7. 729,1000,1331,1728,?
   a) 2137 b)2187 c)2097 d)2197

8. 99, 120, 168, 195, 224, ?
   a) 253 b) 254 c) 255 d) 256

9. 9, 81, 10, 100, 11, 121, 12, 144, 13, 169, ?
   a) 15 b) 225 c) 14 d) 196

10. 1113, 1719, 2329, 3137, 4143, ?
    a) 5152 b) 4096 c) 4753 d) 5159

Find the odd one out among the following :

11. A) Banana B) Mango C) Potato D) Orange

12. A) Cat B) Cow C) Lion D) Sparrow

13. A) Square B) Rectangle C) Triangle D) Sphere

14. A) February B) March C) Sunday D) April

15. A) Blue B) Green C) Yellow D) Sweet

16. A) Pencil B) Eraser C) Chalk D) Paper

17. A) Silver B) Iron C) Copper D) Plastic

18. A) Table B) Bed C) Sofa D) Fan

19. A) Lily B) Lotus C) Jasmine D) Grass

20. A) Moon B) Earth C) Mars D) Star

21. Circus: Entertainment:: Temple: ?
(a) Deity (b) Worship (c) Devotion (d) Donation

22. Proteins: Growth:: Carbohydrates: ?
(a) Energy (b) Strengt (c) Resistance (d) Diseases

23. Bread: Yeast:: Curd: ?
(a) Fungi (b) Bacterial (c) Germs (d) Virus

24. Car: Petrol:: Television: ?
(a) Electricity (b) Transmission (c) Entertainment (d) Antenna

25. College is related to Student in the same way as Hospital is related to,
(a) Doctor (b) Nurse (c) Treatment (d) Patient

26. Man is related to Shout in the same way as Sparrow is related to,
(a)Caw (b) Chirp (c) Mutter (d) Mob

27. Knife: Chopper
(a) Walking: Fitness (b) Swim: Float (c) Scissors: Cloth (d) Quilt: Blanket

28. Birds: Wings
(a) Whale: Water (b) Dog: Lungs (c) Car: Wheel (d) Pen: Paper

29. Rocket: Fuel
(a) Man: Energy (b) Machine: Oil (c) Current: Electricity (d) River: Water

30. East: Orient
(a) North: Polar (b) North: Tropic (c) South: Capricorn (d) West: Occident

KEY :

1-10	b	d	d	c	a	b	d	c	c	a
11-20	c	d	d	c	d	d	d	d	d	d
21-30	b	a	b	a	d	b	d	c	a	d